

Muhammad Jamil Raza Attari

AI Engineer

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SUMMARY

AI/ML Engineer specializing in the end-to-end design and deployment of machine learning systems across Generative AI, Natural Language Processing (NLP), and Computer Vision. Develops advanced LLM applications utilizing Retrieval-Augmented Generation (RAG), LangChain, and vector databases, complemented by model fine-tuning with LoRA. Implements robust MLOps practices, including CI/CD pipelines, model versioning with MLflow, and deploying scalable REST APIs with FastAPI and Docker. Core proficiency in Python, SQL, TensorFlow, PyTorch, Scikit-learn, Transformers, and OpenCV.

EXPERIENCE

AI Engineer

Jan '26 — Present

[Saylani Welfare International Trust](#)

Karachi, Pakistan

Designing and deploying end-to-end AI-powered systems that automate complex business workflows, reduce manual overhead, and scale organizational capabilities. Working across the full stack from LLM integration and agentic pipelines to voice AI and data infrastructure to build production-grade solutions using modern AI tooling.

- 1. Built an AI-powered **Call QA Analyzer** that automatically evaluates inbound call recordings, scores agent performance, and flags quality issues eliminating manual review effort.
- 2. Developed a production **Voice Agent** capable of handling inbound calls autonomously using real-time speech-to-text and LLM reasoning, reducing dependency on human agents.
- 3. Engineered multi-step **AI automation workflows** using n8n, LangChain, and OpenAI/Claude APIs — replacing repetitive manual tasks and saving equivalent effort of multiple FTEs.
- 4. Built and maintained **data pipelines & ETL processes** integrated with SQL databases and external APIs, enabling reliable data flow across AI-powered application

Lead Trainer

Present

SMIT (Saylani Mass IT)

Karachi, Pakistan

Leading hands-on AI/ML training for aspiring engineers covering the full learning journey from Python fundamentals to production ready Agentic AI systems. Focused on bridging the gap between theory and real-world application through structured live instruction and project-based learning

- 1. Delivered live instruction to **30+ students** covering Python, ML fundamentals, Deep Learning, and intro to Large Language Models in a structured bootcamp format.
- 2. Introduced students to **Large Language Models and Agentic AI concepts** covering foundational theory, use cases, and how modern AI agents work, building awareness for real-world AI applications.
- 3. Taught **FastAPI** for deploying AI models as production-ready REST APIs, enabling students to build and ship real-world AI-powered applications end-to-end.
- 4. Designed and delivered **project based curriculum** where students built and presented working AI applications reinforcing concepts through hands-on execution rather than theory alone
- 5. Guided students through the **complete AI development lifecycle** from data preprocessing and model training to deployment building job-ready engineering skills.

ML & AI Engineer

Aug '25 — Present

[Omdena](#)

Remote

Remote

- Built unified preprocessing pipelines fusing **IoT sensor data** with **satellite multispectral imagery** (NDVI/EVI) for crop and irrigation prediction models.
- Engineered tabular feature workflows using scikit-learn **Pipeline** and **ColumnTransformer**, handling missing sensor readings via interpolation and rolling-window imputation.
- Benchmarked **XGBoost** and **scikit-learn** models against **PyTorch / Hugging Face** vision baselines on region-specific agricultural data.
- Mitigated limited ground-truth labels through transfer learning, targeted augmentation, and stratified sampling for sparse regional datasets.
- Collaborated async with a 40+ member global team via Git/GitHub, peer reviews, and weekly sprint demos in an agile workflow.

AI Engineer (Internship)

Oct '25 — Dec '25

[Saylani Welfare International Trust](#)

Karachi, Pakistan

Designed, built, and deployed an end-to-end Retrieval Augmented Generation (RAG) chatbot to automate routine office workflows for medical staff and surface analytical insights for administrators. Owned the complete pipeline document ingestion, embedding, vector retrieval, LLM orchestration, and deployment as the sole developer.

- Built and deployed an end-to-end RAG chatbot that automated routine queries for medical staff, cutting manual lookup time
- Engineered a document-ingestion + embedding pipeline indexing + internal docs for accurate, source-grounded answers.
- Improved answer relevance by tuning chunking and retrieval, validated on a labeled eval set.
- Built an analytics layer surfacing query trends and FAQs for administrators, supporting users.

EDUCATION

Bachelor of Technology, Virtual University

Mar — Present

Intermittade, Government Superior College

Aug '22

SKILLS

Programming Python, SQL, OOP, Async Programming (asyncio), Git / Version Control, Unit Testing (pytest)

GenAI & LLM Engineering LLM Integration (OpenAI / Anthropic APIs), Prompt Engineering, Retrieval Augmented Generation (RAG), AI Agents / Agentic Workflows, Tool / Function Calling, LangChain, LangGraph, OpenAI SDK, Fine-Tuning (LoRA / QLoRA), Structured Outputs (JSON mode / Pydantic), Token & Context-Window Management, Model Context Protocol (MCP)

RAG & Retrieval Systems Document Chunking Strategies, Embedding Models, Semantic Search, Hybrid Search (BM25 + dense), Re-ranking (cross-encoders, Metadata Filtering, Vector Indexing (HNSW / IVF), Query Transformation & Expansion

ML & Deep Learning Foundations TensorFlow / Keras, PyTorch, scikit-learn, Neural Networks, Transformer Architecture, Model Training & Optimization, Hyperparameter Tuning, XGBoost, RAN

Computer Vision OpenCV, Image Classification, Object Detection (YOLO), Image Segmentation, CNN, OCR, Image Preprocessing & Augmentation, Ultralytics

Data Engineering & Pipelines Data Preprocessing & Cleaning, Pandas (Python Package), NumPy, ETL / ELT Pipelines, Feature Engineering, Data Validation (Pydantic), Data Ingestion & Web Scraping

Databases & Vector Stores Vector Databases (Pinecone / FAISS / Qdrant), PostgreSQL, pgvector, Redis, Database Schema Design

Backend & API Development FastAPI, REST API Design, API Authentication (OAuth / JWT), WebSockets / Streaming Responses, Rate Limiting & Retries, Microservices

PROJECTS

Call Center QA System - Saylani Welfare International Trust, Saylani Welfare International Trust

- JSON (audit trail) + SQLite (queryable index) . imageio-ffmpeg . httpx . uv . python-dotenv Engineered an end-to-end AI system to automate call quality assurance for a nonprofit's call center, using an asynchronous FastAPI pipeline to transcribe bilingual (Urdu/English) audio and evaluate agent compliance with
- JSON (audit trail) + SQLite (queryable index) . imageio-ffmpeg . httpx . uv . python-dotenv OpenAI models. The production-ready system features a rule-based scoring engine, duplicate detection, and a Streamlit dashboard for monitoring and managing a manual review queue.

Real-Time Voice AI Ordering System

- Built a full ASR → LLM → TTS speech-to-speech pipeline over WebSocket with hard real-time latency constraints, targeting sub-second Time-to-First-Chunk (TTFC) for conversational voice
- Wired SILMA Arabic TTS with zero-shot voice cloning via reference audio; built a streaming client that yields sentence-level WAV chunks as they arrive to minimize perceived latency
- - Engineered concurrent multi-user architecture using Python ContextVars for per-session state isolation and async SQLite-backed conversation memory across WebSocket sessions

SecureVision AI [Link](#)

- Designed a multimodal AI automation pipeline that monitors Gmail for DVR security events, uses Gemini 2.5 Flash to extract structured alert data from freeform email text, invokes Gemini Vision to generate natural-language scene descriptions from camera frames, and classifies alert severity, delivering enriched WhatsApp notifications via Twilio.

AlphaLens

May '26 — Present

- Building a **serverless RAG agent** that answers natural-language questions over SEC 10-K/10-Q filings with **cited, multi-document synthesis** (~22K chunks, 200 filings).
- Architected a 5-node **single-pass agentic pipeline** in **LangGraph**: plan → hybrid-retrieve → rerank → evaluate → synthesize.
- Designed **hybrid retrieval** — pgvector **HNSW** (768-d) + Postgres **tsvector** full-text, fused via **Reciprocal Rank Fusion**, then **cross-encoder** reranking.

- Built a **dual-backend embedding layer** (Jina v3 API + local nomic-embed fallback) with **quota-based automatic failover**; both emit 768-d vectors for zero-migration switching.

Saylani Qurbani Operations Dashboard, Saylani Welfare International Trust

- Built an end-to-end ETL pipeline in Python/Pandas that ingests 3 years of multi-source data, performs feature engineering (regex-based country inference, price→label mapping, Hijri date conversion) and writes a deduplicated merged dataset via atomic file replace
- Designed and deployed an interactive data visualization dashboard in Streamlit + Plotly with 5 chart modules and gradient KPI scorecards, surfacing donation, branch, and slaughter-funnel insights to non-technical stakeholders.
- Engineered a dual-calendar (Gregorian ↔ Hijri) like-for-like YoY comparison engine that maps any Gregorian window to its Hijri-position equivalent in the prior year — translating a complex business requirement into a reusable analytics function.
- Containerized the application with Docker (Python 3.11-slim) and shipped a one-click in-app pipeline rebuild that refreshes the dataset and clears the Streamlit cache — managed end-to-end via Git/GitHub across 80+ commits

CERTIFICATIONS

AI & Data Science, Saylani Mass IT Training (SMIT)

Sep '25

Agentic AI , Saylani Mass IT Training (SMIT)

May '26